Why carryover erodes power more under covariance moderation than mean moderation: a mechanistic analysis of biomarker-treatment interaction architectures in the Hybrid aggregated N-of-1 design

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Contents

1 Abstract

Background. Simulation is the standard tool for power and sample-size calculation in aggregated N-of-1 trials, designs in which each participant is repeatedly crossed on and off treatment and many single-patient series are pooled in order to validate a predictive biomarker. Any such simulation must specify how the biomarker moderates the treatment effect in its data-generating process (DGP). That specification is a substantive choice rather than an implementation detail, and each of the two established parameterizations carries its own advantages and costs. We formalize two contrasting parameterizations and ask why, in the Hybrid N-of-1 design that motivates this software, carryover erodes power so much more under one architecture than the other.

Methods. We define two DGP architectures. Under 'mean moderation' the biomarker scales the treatment effect in the conditional mean. Under 'covariance moderation' the interaction is represented as a correlation that depends on drug-exposure condition (on drug versus off drug), in a joint multivariate normal (MVN) distribution. Holding the analysis model fixed at a linear mixed model with continuous-time AR(1) residual correlation, we quantify by Monte Carlo simulation the power to detect the interaction as carryover increases in the Hybrid N-of-1 design, the within-subject design used by the prazosin and PTSD reference application that motivates this software. A traditional crossover (CO) and an open-label-plus- blinded-discontinuation design (OL+BDC) are run in parallel, at the same total N, as contrast cases that isolate which design feature drives the architecture-dependent divergence.

Results. Under mean moderation power is nearly preserved in the Hybrid design, because carryover does not disrupt the biomarker's relationship to the treatment effect. At $N = 70$, power falls only from 0.842 to 0.784, a relative loss of 6.9% ($z = 3.34$), as the carryover half-life grows to one week. Under covariance moderation the same carryover erodes the differential correlation that carries the signal, and power falls from 0.840 to 0.711, a relative loss of 15.4% ($z = 2.80$, $p = 0.005$), more than double the mean-moderation loss. The gap traces to a two-channel erosion mechanism specific to covariance moderation: carryover shrinks the drug-exposure contrast, a channel shared with mean moderation, and independently decays the biomarker-response correlation itself, a channel unique to covariance moderation. This second channel's damage scales with how much uninterrupted off-drug time a design exposes patients to, which is why the same architecture contrast is negligible in the classical crossover (CO, $z = 0.29$, where within-subject AR(1) correlation already carries most of the signal covariance moderation depends on) and far larger in the open-label-plus-blinded-discontinuation design (OL+BDC, $z = 7.64$, $p < 10^{-13}$, whose extended blinded-discontinuation phase gives the correlation-decay channel the most uninterrupted time to act). The Hybrid design's intermediate carryover exposure, which alternates on- and off-drug periods rather than sustaining either for long, places its architecture-dependent power loss between these two extremes.

Conclusions. The DGP architecture is a substantive scientific assumption rather than a parameterization convenience, and it determines how severely carryover appears to reduce power. Biomarker-validation simulation studies should state which of the two architectures they adopt and report power under both.

Keywords. N-of-1 trials; biomarker-treatment interaction; data-generating process; multivariate normal; carryover effects; clinical trial simulation; precision medicine; statistical power.

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2 1. Introduction

Precision-medicine trials increasingly ask whether a baseline biomarker modifies the treatment effect, that is, whether a biomarker-treatment interaction is present. For simplified special cases, such as a balanced strict repeated-measures analysis of variance under sphericity, the power of the interaction test does have a closed form, and a companion paper[1] derives it together with its equivalence to a two-sample *t*-test. For the continuous-time mixed model and the N-of-1 designs considered here, however, no such closed form is available, so power is estimated by Monte Carlo simulation. Many trials are generated and analyzed, and the rejection rate is recorded. The simulation must itself specify the biomarker-treatment interaction in its data-generating process, and there is more than one way to do so.

We distinguish two architectures. The common one, direct mean moderation, adds an interaction term to the outcome equation so that the biomarker scales the size of the treatment effect. Essentially every standard trial-simulation framework uses it[2,3]. The second, covariance moderation through differential correlation, used by Hendrickson et al.[4] for N-of-1 trials, instead makes the biomarker change the correlation between treatment response and outcome, so that the interaction is represented in the joint distribution rather than in the mean. On this reading it is the reduced-form second-moment representation of a latent responder-class mechanism, developed at length in a companion paper[5]. Patients may belong to unobserved responder and non-responder groups, with the biomarker only an imperfect indicator of that status, so that two patients with the same biomarker value can still respond differently. Mean moderation cannot represent this situation, because it assumes that the biomarker fixes each patient's drug effect exactly. Covariance moderation represents it by letting biomarker and response track one another only while the drug is acting, as a tendency rather than a fixed rule, a tendency that fades once the drug washes out. As Section 4 shows, that difference is why carryover costs more power under covariance moderation than under mean moderation. A third, 'combined' architecture activates both channels simultaneously through two independent parameters and recovers these two as its single-channel limits. Because it introduces a second free parameter and its power behavior warrants separate treatment, it is developed in a companion paper[6] and is not considered further here.

Choosing between these two architectures is not a matter of computational convenience, because each carries genuine advantages and a genuine cost. Mean moderation is simple, has a single free parameter, and preserves the biomarker-treatment effect ratio under carryover. It commits, however, to a strong assumption, namely that the biomarker deterministically fixes each patient's drug response, an assumption that is implausible whenever the biomarker is only a statistical proxy for an unobserved responder subtype. Covariance moderation relaxes that assumption. As the reduced-form representation of a latent responder-class mechanism discussed above, it lets the biomarker predict response only probabilistically, which is the more defensible choice whenever responder status, rather than the biomarker itself, is the true causal driver, as is plausibly the case for the noradrenergic-tone hypothesis behind the prazosin and PTSD reference application used throughout this paper. That fidelity to the underlying biology comes at a real cost. Because the covariance-moderation signal lives in a correlation between biomarker and response that is sustained only while the drug is active, any off-drug erosion of drug effect directly erodes the statistical signal itself, producing power losses that this paper shows can run many times larger than under mean moderation. Investigators who adopt covariance moderation on biological grounds should therefore budget for its carryover cost at the design stage, rather than discover it after a trial has already been powered.

Hendrickson et al.[4] chose covariance moderation for exactly this reason when they designed the Monte Carlo simulation program behind the prazosin and PTSD trial. Given that trial's own results, covariance moderation was the logical, biologically appropriate specification, because representing the interaction in the mean would have imposed a deterministic biomarker-response relationship that the trial's biology did not support. Mean moderation, by contrast, was and remains the modal specification across the N-of-1 and broader biomarker-simulation literature (Section 4.4), so Hendrickson's choice was a deliberate departure from convention rather than an oversight. What that departure costs, measured in statistical power under carryover, had not previously been quantified, and neither had the mechanism producing that cost been traced to specific, design-relevant features of the trial. Doing both is the central aim of this paper. We take the covariance-moderation choice as scientifically justified for the prazosin and PTSD application, and we ask, in the Hybrid design that trial and this software's target applications use, why carryover erodes power so much more severely under covariance moderation than under mean moderation, so that investigators reasoning by analogy from that trial can budget for the penalty explicitly and understand which design features control its size.

In a single N-of-1 trial one participant is repeatedly crossed between active treatment and a comparator over a sequence of measurement occasions, so that the within-participant contrast between on-drug and off-drug conditions estimates that participant's individual treatment effect. An *aggregated* N-of-1 trial pools many such series in a mixed model to obtain population-level inference at far smaller sample sizes than a parallel-group design. The feature that distinguishes these designs analytically is *carryover*: residual drug effect that decays over days to weeks after discontinuation and contaminates off-drug occasions. Our focus is the Hybrid N-of-1 design, the within-subject design behind the prazosin and PTSD reference application and the design this software was built to support, which alternates on-drug and off-drug periods rather than sustaining either for long. We run two further within-subject designs in parallel as contrast cases that isolate which feature of the Hybrid design's carryover exposure drives its architecture-dependent power loss: a traditional crossover (CO), whose within-subject AR(1) correlation dominates any covariance-moderation signal, and an open-label-plus-blinded- discontinuation design (OL+BDC), whose extended blinded phase gives carryover the most uninterrupted time to act of the three. All three designs are defined in Section 3.

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3 2. Methods

3.1 2.1 Notation and analysis model

Throughout, B_i denotes the biomarker value of participant i, D_{it} the binary on-drug indicator at timepoint t (1 on drug, 0 off drug), and BR_{it} the biological response component of the outcome. These three symbols describe the data-generating-process layer. The analysis-model layer uses a related but distinct continuous drug-exposure variable, `Dbc`, the continuous analogue of D_{it}. It equals 1 during on-drug periods and $\exp(-\lambda t_{sd})$ during off-drug periods, where t_{sd} is time since drug discontinuation and λ is the carryover decay rate, typically derived from the drug's pharmacokinetic half-life as $\lambda = \ln 2/t_{1/2}$. Whereas D_{it} flips abruptly between 0 and 1, `Dbc` captures the smooth shrinkage of residual drug effect during the off-drug period. Code-style identifiers in typewriter font refer to variables in the pmsimstats codebase or to R formula syntax. Mathematical symbols are used otherwise. The analysis model is the same under both architectures, an `nlme::lme` fit with a `corCAR1` continuous-time autocorrelation structure, a random intercept, and a single biomarker-by-drug interaction term, written `bm:Dbc` in R formula syntax. The interaction coefficient on `bm:Dbc` tests whether the biomarker moderates the exposure-decayed drug effect.

3.2 2.2 Two data-generating architectures

3.2.1 2.2.1 Mean moderation

In this architecture the biomarker enters the mean structure of the outcome directly. The treatment effect for participant i at timepoint t is

$$Y_{it} = \mu_0 + \beta_1 \cdot D_{it} \cdot (1 + \beta_{bm} \cdot B_i) + \epsilon_{it},$$

where D_{it} is the on-drug indicator (or a continuous measure of drug exposure), B_i is the participant's biomarker value (typically centered), and β_{bm} is the biomarker moderation parameter. The interaction is explicit, deterministic, and lives in the first moment of the outcome. Participants with higher biomarker values have proportionally larger treatment effects, and this population-level ratio is preserved under any proportional scaling of D_{it}, including the shrinkage that carryover imposes.

This multiplicative form maps directly onto the standard additive regression form for predictive biomarker effects used in the trial methodology literature, specifically the effect-modeling equation of Kent et al.[7], $Y_{it} = \beta_0 + \beta_1 D_{it} + \beta_2 B_i + \beta_3 D_{it} B_i + \epsilon_{it}$, with $\beta_3 = \beta_1 \cdot \beta_{bm}$. The additive form additionally accommodates a prognostic main effect β_2 for the biomarker, which the multiplicative form sets to zero by construction. For power calculations on the interaction coefficient the two forms are interchangeable up to this prognostic-effect distinction, and both fall within mean moderation.

3.2.2 2.2.2 Covariance moderation through differential correlation

In this architecture the biomarker and the biological response (BR) component are jointly drawn from a multivariate normal distribution with a correlation that depends on drug-exposure condition,

$$\begin{pmatrix} B_i \\ BR_{it} \end{pmatrix} \sim \mathrm{MVN}\left(\begin{pmatrix} \mu_B \\ \mu_{BR}(t, D_{it}) \end{pmatrix}, \Sigma(D_{it}) \right), \qquad \mathrm{Cor}(B_i, BR_{it}) = \begin{cases} c_{bm} & D_{it} = 1 \\ c_{bm} \cdot e^{-\lambda t_{sd}} & D_{it} = 0, \end{cases}$$

with the exponential-decay form as the default. The limiting cases $\lambda = 0$ (no decay) and $\lambda \to \infty$ (immediate decay) recover the two no-carryover model variants. The population mean of BR_{it} is the same for all participants with the same treatment history, so the interaction is not in the mean structure. It instead emerges from the conditional distribution. Given $B_i = b$,

$$E[BR_{it} \mid B_i = b, D_{it} = 1] = \mu_{BR}(t) + c_{bm} \cdot \frac{\sigma_{BR}}{\sigma_B} \cdot (b - \mu_B).$$

The interaction here is probabilistic rather than deterministic, lives in the second moment of the joint distribution, and persists with attenuated strength off drug, vanishing only in the limit of complete decay. Carryover in the DGP inflates the BR means during off-drug periods, making them resemble on-drug means. This reduces the differential in the correlation structure between drug-exposure conditions. Section 2.2.3 shows that this erosion leaves the analysis model's point estimate of the interaction largely unbiased, but it converts variance the biomarker previously explained into unexplained residual noise, which is the channel through which it costs power (Section 3.2). A third, combined architecture, in which both channels operate at once through independent weights, recovers the two architectures above as its single-channel limits and is the natural framework when the operative mechanism is uncertain. It is developed in a companion paper[6].

3.2.3 2.2.3 Analytical comparison

Consider the expected outcome difference between two participants with biomarker values b_1 and b_2 at the same drug exposure level D. Under mean moderation,

$$E[Y \mid b_1, D] - E[Y \mid b_2, D] = \beta_1 \cdot \beta_{bm} \cdot (b_1 - b_2) \cdot D.$$

This scales linearly with D and therefore shrinks during off-drug periods, when D is replaced by an exposure-decayed value. However, the *ratio* of drug effect between high- and low-biomarker participants is preserved at every level of exposure, so the biomarker-treatment signal contracts in amplitude but does not lose identifiability under carryover. Under covariance moderation the analogous quantity, during off-drug periods with carryover, is

$$E[BR \mid b, D=0, t_{sd}] = \mu_{BR,\text{off}}(t_{sd}) + c_{bm} \cdot e^{-\lambda t_{sd}} \cdot \frac{\sigma_{BR}}{\sigma_B} \cdot (b - \mu_B).$$

Here the biomarker-dependent component decays exponentially with time off drug. As t_{sd} increases, the off-drug conditional expectation converges to $\mu_{BR,\text{off}}$ for all biomarker values, in contrast to mean moderation's preserved ratio.

This convergence in the raw time since discontinuation is not, on its own, the source of the power loss documented in Section 3, and disentangling the two requires re-expressing the comparison in terms of the regressor the analysis model actually uses. Write the exponential decay factor as $\kappa_{it} = 1$ on drug and $\kappa_{it} = e^{-\lambda t_{sd,it}}$ off drug, matching the definition of $D_{bc,it}$ in Section 2.1 exactly when the analysis-model half-life is set equal to the DGP half-life, the base case examined in Section 3.1. Then $\text{Cor}(B_i, BR_{it}) = c_{bm} \cdot D_{bc,it}$ by construction, and the conditional mean above is *exactly linear* in $D_{bc,it} \cdot (b - \mu_B)$ at every value of t_{sd}, not only in the untouched limit:

$$E[BR_{it} \mid B_i = b, D_{it}] = \mu_{BR}(t) + c_{bm}\, D_{bc,it}\, \frac{\sigma_{BR}}{\sigma_B}(b - \mu_B).$$

Because the analysis model regresses the outcome on `bm:Dbc` using this same $D_{bc,it}$, its population (large-sample) estimate of the interaction slope recovers $c_{bm}\, \sigma_{BR}/\sigma_B$ *regardless of how much carryover has occurred*, to the same leading order that mean moderation's slope $\beta_1\beta_{bm}$ is carryover-invariant. The biomarker-dependent signal is not eliminated by carryover in the sense that matters for the fitted interaction coefficient; only its raw amplitude at long t_{sd} is. This is confirmed empirically in Section 3.4: the mean of $\hat{\beta}_{bm:D_{bc}}$ under covariance moderation holds near -0.234 across every carryover level examined, rather than trending toward zero, exactly as this identity predicts.

If the population slope is unaffected, the power loss under covariance moderation must arise instead from the precision with which that slope is estimated, and the bivariate normal structure of Section 2.2.2 gives that precision loss in closed form. Conditional on B_i, the variance of BR_{it} is

$$\text{Var}(BR_{it} \mid B_i, D_{it}) = \sigma^2_{BR}(1 - c^2_{bm}\, D^2_{bc,it}),$$

an exact consequence of the bivariate normal conditional-variance identity applied to a correlation of $c_{bm}\, D_{bc,it}$. On drug ($D_{bc,it} = 1$), the biomarker explains a fixed c^2_{bm} fraction of BR's variance; as carryover erodes $D_{bc,it}$ toward zero off drug, that explained fraction shrinks *quadratically*, not linearly, in $D_{bc,it}$, and the corresponding conditional variance rises toward the full unconditional σ^2_{BR}, adding pure noise that the fixed-effects analysis model, which assumes homoscedastic residual variance, cannot distinguish from measurement error. Mean moderation has no analogue of this channel: its residual variance ϵ_{it} is constant by construction (Section 2.2.1), independent of D_{it}, so carryover there degrades only the regressor's contrast, never the noise floor around it. This is the quantitative content behind the qualitative two-channel account of Section 3.2: mean moderation loses information only through the design contrast in $D_{bc,it}$ itself, a channel shared by both architectures, while covariance moderation loses information through this quadratic variance-inflation channel as well, and it is this second, architecture-specific channel that produces the divergence in power documented in Section 3.

3.2.4 2.2.4 Relationship to Hendrickson et al.'s original implementation

Covariance moderation, as formalized in Section 2.2.2, is a corrected adaptation of Hendrickson et al.'s own published mechanism, not an identical reproduction of it. We verified this directly against Hendrickson et al.'s original public repository (`github.com/rchendrickson/pmsimstats`, the commit history closest to their 2020 acceptance date) rather than against a description of it, and found three concrete differences, together with a fourth consequence of the second that bears directly on this paper's choice of specification.

First, Hendrickson et al.'s original data-generating process builds within-factor and cross-factor correlation from a single constant applied to every pair of timepoints regardless of separation, compound symmetry, whereas the architectures examined here use AR(1) decay, $\mathrm{Cor}(X_{t_i}, X_{t_j}) = \rho^{|t_i - t_j|}$, which we adopt for its greater clinical plausibility (nearby measurements should be more correlated than distant ones) as well as the numerical benefit described next. Second, the biomarker-BR correlation itself: inspection of the original `generateData.R` shows that Hendrickson et al.'s code does directly parameterize a drug-state-dependent Cor(BM, BR), discounted off drug by the same exponential decay applied to the BR mean, so the mechanism formalized in Section 2.2.2 is a faithful generalization of theirs, not a reinterpretation of an ambiguous original; the one difference is an additional, undocumented `scalefactor` constant (defaulting to 2) multiplying t_{sd} in the decay exponent, which we omit as unmotivated by any stated pharmacokinetic rationale. Third, and independently of the DGP, Hendrickson et al.'s own reproducibility vignette (`Produce_Publication_Results_1_generate_data.Rmd`, confirmed by direct inspection) sets the analysis model's `t_random_slope` option to `FALSE` for the published results, giving `lme4::lmer` with a random intercept only and no explicit residual correlation structure; this contradicts their own Methods section, which describes "maximal random effects structure…an unstructured variance-covariance matrix." We adopt `nlme::lme` with `corCAR1` (Section 2.1) instead, consistent with the AR(1) DGP correction, because a model that ignores residual autocorrelation under an autocorrelated DGP is misspecified in a way that is already known to inflate Type~I error, a defect independent of, and additional to, the architecture question this paper addresses.

The AR(1) correction has a direct numerical consequence for the range of c_{bm} values this paper's simulations can explore. Compound symmetry restricts how large a biomarker-BR correlation the covariance matrix can support before it fails to be positive definite, because a constant correlation applied uniformly across many timepoints depletes the matrix's degrees of freedom faster than a correlation that decays with lag. We confirmed this directly: holding every other parameter at this paper's calibration (OL+BDC design, $t_{1/2} = 1.0$), the raw, uncorrected covariance matrix under Hendrickson et al.'s original compound-symmetry construction loses positive definiteness beyond $c_{bm} \approx 0.35$ and is already non-positive-definite at $c_{bm} = 0.45$, the value used throughout Section 3; under the AR(1) construction used here, the same matrix remains positive definite up to $c_{bm} \approx 0.50$, comfortably covering $c_{bm} = 0.45$ without requiring the positive-definiteness correction step at all. This matches, and sharpens to this paper's own specific configuration, the general finding reported elsewhere in this project that AR(1) expands the positive-definite-feasible region of c_{bm} from a compound-symmetry maximum near 0.25 to 0.49 or higher. Every cell in Section 3 that uses $c_{bm} = 0.45$ would therefore require `corpcor::make.positive.definite()` to silently force Hendrickson et al.'s original covariance matrix onto the nearest positive-definite matrix before every single simulated draw, a correction this paper's own architectures never need to invoke at this calibration.

To characterize what these three differences do to the reported magnitudes, we ran a fourth, non-primary simulation arm: Hendrickson et al.'s original mechanism exactly as published (their compound-symmetry DGP with the default `scalefactor_2` decay, and their own random-intercept-only `lmer` analysis, including its use of a fixed, DGP-independent Dbc treatment, since their reproducibility code never passes the DGP's true $carryover\_t1half$ into the analysis step) through the same three designs, carryover half-lives, and $c_{bm} = 0.45$ used in Section 3.1, at 300 replicates per cell (versus this paper's production 1{,}000; MCSE on power near 0.5 is approximately 0.029). The relative loss from $t_{1/2} = 0$ to $t_{1/2} = 1.0$ was 4.2% (CO), 12.5% (Hybrid), and 19.0% (OL+BDC), the identical qualitative ordering reported for covariance moderation in Section 3.1, but smaller in magnitude than this paper's Architecture B at every design (3.0%, 15.4%, and 30.6% respectively) and larger than Architecture A at every design (1.9%, 6.9%, 2.8%). Hendrickson et al.'s original combination therefore sits between the two architectures examined in this paper, not on top of either: the covariance-based mechanism common to their DGP and Architecture B drives the same design-dependent ordering in both, while the three corrections, compounded, materially change its size, most visibly in OL+BDC, where this paper's matched-half-life `corCAR1` analysis reports over 1.6 times the relative loss Hendrickson et al.'s own original code would report on the same designs and calibration. This result is reported for context on the sensitivity of the reported magnitudes to implementation choices, not as a claim about which published figure is more correct; Hendrickson et al.'s own carryover grid (0, 0.1, 0.2 weeks) and step-function DGP decay differ from the smooth exponential decay and 0-1.0 week range used throughout this paper, so no single number from either implementation is directly comparable to Hendrickson et al.'s published Figure 4B.

3.3 2.3 Simulation design

We ran identical simulation configurations under both architectures using the pmsimstats-ng framework, matching the DGP parameters as closely as possible. The non-biologic response components (time-varying response, placebo-belief accumulation, and residual noise) are generated from the same multivariate normal draw, with identical means, variances, within-factor AR(1) autocorrelation, and cross-factor correlations, under both architectures. The two DGPs differ only in how the biomarker-to-biological-response linkage is specified, so the divergent carryover behavior reported in Section 3 is attributable to that parameterization alone and not to any difference in how the other components are generated.

We follow the ADEMP framework of Morris, White, and Crowther[8], comprising Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures. The Aim is to compare power to detect the biomarker-by-drug interaction across architectures as carryover increases. The Data-generating mechanisms are the two architectures of Section 2.2. The **primary estimand** is the interaction coefficient $\beta_{bm:D}$ in the model `Sx ~ bm + t + Dbc + bm:Dbc`. The **Methods** setup is as follows.

- Trial designs: CO (traditional crossover), Hybrid (N-of-1 Hybrid), OL+BDC (open-label with blinded discontinuation).
- Sample size: $N = 70$ in total for every design, allocated across each design's randomization paths, so the three designs are matched on N. The two-path CO and OL+BDC designs put 35 on each path, and the four-path Hybrid design splits 70 as 18/18/17/17.
- Biomarker correlation or moderation: $c_{bm} = 0.45$, equivalently $\beta_{bm} = 0.45$.
- DGP carryover half-life: $t_{1/2} \in \{0, 0.5, 1.0\}$ weeks, with the analysis-model `Dbc` half-life matched to it.
- Replicates: 1,000 per cell (Monte Carlo standard error, MCSE, on a binomial power estimate near 0.5 is approximately 0.016).
- Implementation: 8-core `parallel::mclapply`, full factorial of 2 architectures $\times$ 3 designs $\times$ 3 $t_{1/2}$ $\times$ 2 c_{bm} levels gives 36 cells and 36,000 fits, with 100% convergence.

The **Performance measures** are power for $\beta_{bm:D}$ at $\alpha = 0.05$, Type I error under $c_{bm} = 0$ (both with MCSE $\sqrt{\hat{\pi}(1-\hat{\pi})/n_{\text{reps}}}$), bias of $\hat{\beta}_{bm:D}$ relative to the reference true value $\theta_{\text{true}} = -c_{bm} \cdot \sigma_{BR}$, the same calibration used in the companion carryover-specification and component-decomposition manuscripts[9,10], the empirical MCSE of $\hat{\beta}_{bm:D}$ across replicates, and the convergence rate. The number of replicates per cell is chosen to achieve MCSE below 0.02 for power at $\hat{\pi} = 0.5$, which corresponds to $n_{\text{reps}} \geq 625$. We use $n_{\text{reps}} = 1000$ throughout.

The $N = 70$ choice is the smallest of the prazosin-PTSD-matched sample sizes used in the published methodological literature, and it is a regime in which no design or architecture saturates power at the no-carryover cells (the highest cell in the grid is 0.842), so carryover-induced erosion is mechanically visible. This is the main practical gain from matching the designs on N. At a previous per-path parameterization the Hybrid cells sat above 0.99, where no erosion could register.

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4 3. Results

Covariance moderation, as Section 1 argues, is often the more biologically defensible choice when a biomarker is best understood as predictive rather than causal, and it is the specification Hendrickson et al. selected for the prazosin and PTSD application on exactly these grounds. That defensibility does not eliminate a carryover cost. Quantifying that cost in the Hybrid design, and tracing why it arises, so that the appeal of the architecture does not obscure a real design consideration, is the purpose of what follows. The crossover (CO) and open-label-plus-blinded- discontinuation (OL+BDC) designs are reported alongside Hybrid at every step, not as objects of primary interest, but because contrasting Hybrid against these two structurally different designs is what identifies which feature of a design's carryover exposure drives the architecture-dependent loss.

4.1 3.1 Power under carryover at $N = 70$

Mean moderation.

Design	$t_{1/2} = 0$	$t_{1/2} = 0.5$	$t_{1/2} = 1.0$
CO	0.739	0.739	0.725
Hybrid	0.842	0.844	0.784
OL+BDC	0.751	0.748	0.730

Power declines modestly under mean moderation. In the Hybrid design, the relative loss from $t_{1/2} = 0$ to $t_{1/2} = 1$ is 6.9%, and it is the only one of the three designs whose loss is separable from zero at 1{,}000 replicates per cell ($z = 3.34$). The two-period designs lose less (1.9%, CO; 2.8%, OL+BDC) and neither loss is distinguishable from Monte Carlo noise (CO $z = 0.71$; OL+BDC $z = 1.07$). All three values fall at or below the 8-13% relative-loss range suggested by earlier exploratory runs in this project for direct-mean-moderation DGPs, an erosion that leaves power substantially intact even in the design, Hybrid, where it is largest.

Covariance moderation (differential correlation, MVN).

Design	$t_{1/2} = 0$	$t_{1/2} = 0.5$	$t_{1/2} = 1.0$
CO	0.745	0.754	0.723
Hybrid	0.840	0.795	0.711
OL+BDC	0.777	0.694	0.539

Power declines more sharply under covariance moderation than under mean moderation in every design, but by very different amounts. In the Hybrid design, the relative loss from $t_{1/2} = 0$ to $t_{1/2} = 1$ is 15.4%, more than double its matched mean-moderation loss (6.9%). The two-period designs bracket this value from opposite sides: CO loses only 3.0% (not significantly different from mean moderation's CO loss), while OL+BDC loses **30.6%**, roughly 11 times its matched mean-moderation loss (2.8%) and twice the Hybrid loss. The CO cell shows a small, non-monotone fluctuation ($0.745 \to 0.754 \to 0.723$), within one MCSE and consistent with no architecture-by-carryover interaction in that design. These figures fall below the 40-60%

relative-loss range suggested by earlier exploratory runs for the MVN differential-correlation DGP, but confirm their qualitative finding of substantial, design-dependent erosion, with the Hybrid design occupying a genuine middle position rather than resembling either contrast case.

We compare the carryover-induced power loss between architectures using a difference-of-differences z-statistic. Let $\Delta_X = \hat{\pi}_{X,0} - \hat{\pi}_{X,1}$ denote the within-architecture loss from $t_{1/2} = 0$ to $t_{1/2} = 1.0$ for architecture X, estimated from $n = 1{,}000$ independent replicates per cell. The covariance-minus-mean gap $\Delta_B - \Delta_A$ has standard error

$$\mathrm{SE} = \sqrt{\frac{\hat{\pi}_{B,0}(1-\hat{\pi}_{B,0})}{n} + \frac{\hat{\pi}_{B,1}(1-\hat{\pi}_{B,1})}{n} + \frac{\hat{\pi}_{A,0}(1-\hat{\pi}_{A,0})}{n} + \frac{\hat{\pi}_{A,1}(1-\hat{\pi}_{A,1})}{n}},$$

since the four proportions come from independent replicate sets (all tests two-sided). In the Hybrid design, $\Delta_A = 0.842 - 0.784 = 0.058$ and $\Delta_B = 0.840 - 0.711 = 0.129$, giving $\Delta_B - \Delta_A = 0.071$ (about 7 percentage points) and $z = 2.80$ ($p = 0.005$): the architectures' carryover behavior differs well beyond Monte Carlo error even in this intermediate design. The same contrast is much larger for OL+BDC ($\Delta_A = 0.021$, $\Delta_B = 0.238$, $\Delta_B - \Delta_A = 0.217$, about 22 percentage points, $z = 7.64$, $p < 10^{-13}$) and indistinguishable from zero for CO ($z = 0.29$), because the CO design's within-subject AR(1) structure dominates the carryover-mediated correlation-decay channel that covariance moderation uses to represent the interaction, leaving little architecture-dependent signal for the contrast to detect. The ordering of the three designs on this contrast, OL+BDC well above Hybrid and Hybrid above CO, is the substantive result, and it is consistent with a single explanatory factor: the amount of uninterrupted off-drug time each design's structure exposes patients to. It is read at a common $N = 70$ and is not a statement about the designs' relative power (Section 4.1).

4.2 3.2 Mechanism: why the architectures diverge

Section 2.2.3 identified two channels through which carryover degrades the interaction signal, and only one of them is shared by both architectures. **Design-contrast erosion** shrinks $\mathrm{Var}(D_{bc,it})$ as carryover pulls off-drug exposure values away from a clean 0/1 contrast; because the interaction regressor is `bm:Dbc`, less contrast in `Dbc` means less information for the interaction coefficient under either architecture, and this channel alone explains mean moderation's modest, design-dependent loss (1.9-6.9%). **Conditional-variance inflation** is unique to covariance moderation and is the channel that produces the much larger losses of Section 3.1. As $D_{bc,it} \to 0$ off drug, $\mathrm{Var}(BR_{it} \mid B_i, D_{it}) = \sigma^2_{BR}(1 - c^2_{bm} D^2_{bc,it})$ rises from its on-drug floor of $\sigma^2_{BR}(1 - c^2_{bm})$ toward the full unconditional σ^2_{BR}, converting variance that was explained by the biomarker into unexplained residual noise that a homoscedastic fixed-effects model cannot separate from measurement error. Mean moderation has no such channel: its residual variance is constant by construction, so it loses only design contrast, never precision around a fixed contrast. This is why the population interaction coefficient stays essentially unbiased under both architectures (Section 3.3) while power diverges sharply: the loss is a precision effect specific to covariance moderation, not a bias effect shared by both.

A rough, working-independence approximation to the interaction test's noncentrality parameter makes the connection to design and carryover explicit. Treating each observation's contribution to the Wald statistic as approximately additive and ignoring the `corCAR1` residual correlation that the actual analysis model estimates, the information available from an observation at exposure level $D_{bc,it} = x$, after partialling out the model's own `Dbc` main effect, is approximately proportional to $(x - \bar{D}_{bc})^2/\sigma^2_\epsilon$ under mean moderation and to $(x-\bar{D}_{bc})^2/[\sigma^2_{BR}(1-c^2_{bm}x^2)]$ under covariance moderation, where $\bar{D}_{bc}$ is the design's mean exposure level. Both summands shrink as carryover compresses $D_{bc,it}$ toward $\bar{D}_{bc}$ across the design, and the covariance-moderation summand is additionally discounted by the same conditional-variance-inflation factor derived above, so the total information lost accumulates fastest in designs that spend the most observation-weeks away from a clean on/off contrast.

This approximation was evaluated numerically against each of the three designs' exact time-point and phase schedules (constructed with the package's own `buildtrialdesign()`, at $c_{bm} = 0.45$), rather than left as an unexecuted next step, and the result is mixed. Summing $(D_{bc,it} - \bar{D}_{bc})^2$ over every observation in each design correctly orders the three by predicted information loss from $t_{1/2} = 0$ to $t_{1/2} = 1.0$: 36.8% in the focal Hybrid design, against 4.6% in CO and 51.2% in OL+BDC, matching the qualitative ordering of Section 3.1's power losses under either architecture. But the covariance-moderation summand predicts an architecture-specific gap over mean moderation's own loss of under one percentage point at every design (37.1% in Hybrid, 4.7% in CO, 50.9% in OL+BDC) at $c_{bm} = 0.45$, far smaller than the large gap Section 3.1 actually reports (for example, Hybrid's 6.9% mean-moderation loss against its 15.4% covariance-moderation loss, or OL+BDC's 2.8% against 30.6%). The exact conditional-variance-inflation identity itself is not in question here; it is independently confirmed against realized Monte Carlo variability in Section 3.3. What this numeric check shows is that the working-independence simplification used to connect that identity to design and carryover analytically discards enough of the `corCAR1` structure to badly underpredict the size of the architecture gap, not only its near-absence in CO. A full analytical account of the magnitude, rather than only its direction, would need to incorporate the within-subject autocorrelation directly, which remains undone; the qualitative account below is offered as the mechanism for CO's particular robustness in place of that missing derivation.

Qualitatively, the ordering of Section 3.1 follows the same information argument applied to each design's own pattern of drug-exposure alternation. Conditional-variance inflation needs sustained low-$D_{bc,it}$ time to accumulate: the longer a design lets $D_{bc,it}$ sit near zero before the next on-drug period resets it, the more of that period's information is converted to noise. OL+BDC sustains its off-drug window the longest of the three designs, so its variance-inflation channel does the most damage. CO alternates quickly enough, and leans on within-subject AR(1) correlation heavily enough to have already extracted most of the available signal by other means, that little conditional-variance inflation has room to act before the next on-drug period arrives. The Hybrid design's paths mix on- and off-drug periods at an intermediate cadence, so $D_{bc,it}$ rarely stays near zero as long as it does under OL+BDC, capping how much variance inflation any single off-drug stretch can accumulate; the design's intermediate architecture-dependent loss (Section 3.1) is the empirical signature of this intermediate exposure pattern.

4.3 3.3 Type I error and estimator bias

Across all 18 null cells of the Section 3.1 production run, Type I error sat in $[0.010, 0.074]$, with mean 0.042 and median 0.035, and no architecture-specific inflation. The uniformly sub-nominal rates are consistent with the `corCAR1` working-covariance miscalibration characterized in a companion calibration study[11], whose model-based standard error is over-stated by roughly 6 to 10 percent. Because this affects both architectures alike, it is common-mode with respect to the architecture contrast and does not affect it.

Mean estimated $\hat{\beta}_{bm:D_{bc}}$ at $c_{bm} = 0.45$ ranged from -0.230 to -0.281 across the 18 alternative cells, with the mean-moderation OL+BDC cell at $t_{1/2} = 1.0$ producing the largest absolute mean. Covariance-moderation cells maintained a mean near -0.234 across all $t_{1/2}$ values, exactly as the unbiasedness argument of Section 2.2.3 predicts: the population interaction slope recovered from $D_{bc,it}$ does not depend on carryover once the analysis model's half-life matches the DGP's. Empirical SE of $\hat{\beta}_{bm:D_{bc}}$ runs 0.07 to 0.10 across cells, and the within-replicate model SE matches it at scale, which is consistent with nominal-95% Wald coverage but was not directly verified by a coverage simulation.

The conditional-variance-inflation mechanism of Section 3.2 can be checked directly against this same table, since it predicts where the empirical SE should grow with carryover and where it should not. Computing $|\hat{\beta}_{bm:D_{bc}}|/\mathrm{SE}(\hat{\beta}_{bm:D_{bc}})$ at $c_{bm} = 0.45$ as a design- and architecture-specific proxy for the interaction test's signal-to-noise ratio, and comparing its value at $t_{1/2} = 0$ against $t_{1/2} = 1.0$: under covariance moderation this ratio falls by 12.2% in OL+BDC (2.947 to 2.588) and 12.6% in Hybrid (3.259 to 2.849), while under mean moderation it falls by only 5.4% in Hybrid (3.002 to 2.839) and is flat to slightly higher in OL+BDC (-2.7%, 2.865 to 2.943) and in CO (0.6%, mvn; 1.9%, mean moderation). This is the pattern Section 3.2 predicts: covariance moderation's SNR erosion is driven almost entirely by SE growth (0.079 to 0.090 in OL+BDC, a 14.6% increase) against an essentially flat numerator, consistent with the derived $\sigma^2_{BR}(1 - c^2_{bm} D^2_{bc,it})$ variance-inflation identity, whereas mean moderation shows no comparable SE trend because its residual variance does not depend on $D_{bc,it}$. This check is verified directly from the committed cell-level summaries (`01-dgp-summary.txt`) rather than from a new simulation, and it substantiates the mechanism of Section 3.2 quantitatively, though it remains a post hoc check on realized Monte Carlo variability rather than a derivation of the noncentrality parameter from design and DGP parameters alone.

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5 4. Discussion

5.1 4.1 Matching architecture to biological mechanism

The results of Section 3 reduce to a single trade-off that this subsection develops at length. Mean moderation is simpler, is essentially immune to carryover, and is directly comparable to the overwhelming majority of the published biomarker-simulation literature, but it assumes a mechanism that is often biologically wrong: that the biomarker fixes each patient's drug response exactly, rather than merely predicting it. Covariance moderation is biologically faithful in exactly the cases where mean moderation is not, because it lets the biomarker predict response only probabilistically, but that fidelity is purchased at a real, design-dependent power cost under carryover, a cost of the kind that an investigator who ran their power calculation under mean moderation alone would never see in their own simulation and would therefore not budget for. Neither architecture dominates the other in the abstract; the correct choice, and the size of the risk from choosing wrong, depends on the biology of the specific biomarker under study, a claim this subsection substantiates first from first principles and then from the finite-mixture formalism developed at length in a companion paper.

The choice of architecture is a choice about *why* the biomarker predicts response, and is therefore a biological statement rather than a coding preference. Mean moderation is appropriate when the biomarker mechanically sets the size of the drug effect, for example by fixing effective dose or receptor occupancy, so that doubling the biomarker roughly doubles the effect at every timepoint. Because carryover keeps the same proportion off drug, power is barely touched. Typical examples include drug clearance (kidney function setting the steady-state level of a renally cleared drug), receptor density, and metabolizer genotype. Covariance moderation is appropriate when the biomarker is only a proxy for a hidden responder status that it indexes imperfectly, so that identical biomarker values can still respond differently, and the biomarker predicts response only while the drug is acting. Once the drug washes out, the correlation that carried the signal fades; the analysis model's point estimate stays close to unbiased even so (Section 2.2.3), but the precision with which it can be estimated drops, and power falls sharply as a result (Section 3.2). Typical examples include resting blood pressure as a marker of noradrenergic tone, inflammatory markers such as CRP or IL-6, and polygenic risk scores. A fully faithful model of the hidden responder-class idea would be a finite mixture. Covariance moderation is a convenient correlation-level approximation, adequate for power calculations but not a claim about the true mechanism. A companion paper[5] develops the mixture and latent-class formulations, deriving when a single multivariate normal reproduces a two-component mixture's first two moments and cataloguing where it fails. Senn[12] also cautions that apparent biomarker-treatment effects are sometimes just variance artifacts, so covariance moderation's signal may be smaller than assumed.

The pmsimstats software was built for a prazosin/PTSD trial[4,13], in which the biomarker, resting blood pressure, is thought to be a proxy for noradrenergic tone rather than a direct driver of drug levels, a covariance-moderation story. The carryover-driven power loss we find under covariance moderation is therefore a clinically relevant warning. Designs with long off-drug periods may lose more power to detect this biomarker than a mean-moderation simulation would suggest. Should blood pressure act through the mean channel as well as the correlation channel, the dual-channel architecture of the companion paper[6] is the appropriate model, and because it estimates two independent parameters, it calls for independent evidence on each.

Three further considerations support covariance moderation as the more defensible specification for this application. First, resting blood pressure is a single-occasion vital sign subject to substantial physiological and measurement variability, so treating it as a precise multiplicative scale factor on the treatment effect assumes a level of measurement precision the biomarker does not have. A specification that asserts only an association, and its strength, is proportionate to that measurement error. Second, the distinction tracks an established one in the predictive-biomarker literature. The PATH framework of Kent et al.[7] separates a causal treatment effect modifier from a biomarker that is merely predictive of response without being causally upstream of it. Covariance moderation is the natural representation of the predictive role and mean moderation of the causal-mediator role, so classifying blood pressure as predictive rather than causal in this trial argues for covariance moderation on grounds already established in that literature. Third, mean moderation requires committing to a specific functional form for how the biomarker scales the treatment effect, typically linear in the centered biomarker. Covariance moderation makes the weaker assumption that biomarker and response are associated, without specifying the functional form of that association beyond its strength, which is the more parsimonious choice when investigators lack direct evidence for a particular dose-response shape.

The two architectures considered here are the two poles of a wider spectrum, and drug and biomarker properties can place a candidate application closer to one boundary of that spectrum than to either pole outright. A companion paper on latent-class and mixture-model formulations[5] develops this spectrum at length and grounds covariance moderation as a specific, derivable approximation to a genuinely latent-class generative mechanism, an argument summarized here because it is the strongest available justification for covariance moderation's biological faithfulness and, at the same time, the source of its principal limitation.

The mixture argument begins from the observation that if a candidate biomarker indexes an unobserved responder or non-responder dichotomy, as the noradrenergic-tone hypothesis behind the prazosin and PTSD reference application supposes, the mechanistically faithful DGP is not a correlation but a finite mixture,

$$Z_i \mid B_i \sim \mathrm{Bernoulli}(\pi(B_i)), \qquad BR_{it} \mid Z_i = z \sim f_z(t, D_{bc,it}), \qquad \pi(B_i) = \mathrm{expit}(\psi_0 + \psi_g b_i),$$

where Z_i is an unobserved class label fixed for participant i across the whole trial, b_i is the standardized biomarker, and the gating slope ψ_g governs how steeply the biomarker predicts class membership. Two properties of this generative form matter for what follows. First, Z_i is invariant across drug state and time: once a participant is drawn into class 1 or class 0, that membership persists through every on-drug and off-drug measurement occasion in the trial, in principle recoverable from the full within-subject trajectory even at timepoints where the instantaneous drug-exposure contrast has decayed to near zero. Second, the class-conditional mean trajectory itself still depends on $D_{bc,it}$, so a full carryover off-drug period drives the two class-conditional means toward each other if the responder and non-responder distinction lies purely in drug response, exactly the mean-blurring channel that Section 3.2 identifies as common to both architectures examined here.

Under within-class independence between B_i and BR_{it}, a two-component mixture of this form induces a marginal joint distribution of (B_i, BR_{it}) whose first two moments are reproduced, to leading order, by a single multivariate normal with $\mathrm{Cor}(B, BR) = c_{bm}$, where

$$c_{bm}^2 = \frac{\pi(1-\pi)(\mu_B^1 - \mu_B^0)^2}{\mathrm{Var}(B)} \cdot \frac{\pi(1-\pi)(\mu_{BR}^1 - \mu_{BR}^0)^2}{\mathrm{Var}(BR)},$$

the product of the between-class variance fractions in the biomarker and in the biological-response component. Under mild regularity conditions (adequate class overlap, mixing proportion π bounded away from 0 and 1), this second-moment structure carries a differential correlation across drug-exposure conditions, elevated where the class-conditional means separate on drug and shrinking toward zero where they coincide off drug after carryover has run its course, which is precisely the covariance-moderation DGP of Section 2.2.2. Covariance moderation is therefore not an arbitrary alternative to mean moderation; it is the specific second-moment approximation to a latent-class mechanism that a biomarker acting as an imperfect proxy for a hidden responder subtype would actually generate, which is the formal grounding for treating it as the more biologically faithful choice whenever that subtype story is plausible.

That derivation also exposes covariance moderation's principal limitation directly. An MVN approximation reproduces only the first two moments of the mixture; it discards the class label Z_i itself, along with the class-invariance property that made Z_i carryover-robust in the true generative form. A correlation, unlike a class label, must be estimated fresh from the currently observed joint distribution at every timepoint, with no persistent per-participant memory once off-drug attenuation has taken effect. This is why the power losses documented in Section 3 are a property of the covariance-moderation *approximation* specifically, not an inherent, unrecoverable cost of a genuinely latent-class biology. A class-aware analysis fitted to mixture-DGP data could, in principle, recover some of that apparent loss by using the carryover-invariant class label directly, up to an asymptotic bound set by the gap between the covariance-only and mixture-faithful power estimates. In practice, though, that recovery is bounded by identifiability: fitting a finite mixture (an EM algorithm, or its implementations in `lcmm` or `flexmix`) reliably distinguishes genuine class structure from noise only at sample sizes of several hundred to several thousand in the psychometric literature that developed these methods[14–16], well above the $N \in [30, 150]$ typical of N-of-1 biomarker-validation trials. At trial-relevant N, mixture analysis may be underpowered or unidentified even when the underlying biology is genuinely mixture-structured, so a class-aware primary analysis is not generally a usable alternative to the standard mixed-model analysis of Section 2.1, only a sensitivity check where identifiability permits. Covariance moderation is therefore best understood as the pragmatic compromise this identifiability constraint forces: the representation that is faithful to the underlying biology at the level of the joint distribution's first two moments, without requiring an estimation strategy that trial-relevant sample sizes cannot support. When the biomarker is a near-deterministic class indicator, for example a genotype with near-perfect sensitivity and specificity for a metabolizing versus non-metabolizing phenotype, the class-membership gradient is steep and a thresholded mean-moderation specification is the better approximation even though the underlying mechanism is a latent subtype rather than a graded scaling. When the two response classes are well separated, as in some targeted-oncology indications with a driver mutation that essentially abolishes response in its absence, the marginal response distribution is genuinely bimodal and neither architecture is adequate: a full mixture DGP, not examined by simulation in this paper, is required to avoid understating the power achievable by a class-aware analysis. When covariance, autocorrelation, or placebo-response parameters differ by latent class rather than only the mean or the biomarker correlation, a single-component MVN of either architecture is misspecified regardless of which moment carries the interaction. Investigators should therefore treat mean moderation and covariance moderation as convenient limiting cases to be checked against, rather than an exhaustive pair of options, whenever pilot data suggest the biomarker may index a genuinely discrete or highly separated subgroup structure.

The trial-design dependence documented in Section 3, largest in OL+BDC and smallest in CO, is the empirical signature of the same class-label discarding just established: because covariance moderation carries no persistent per-participant memory of responder status once off-drug attenuation has run its course, the signal it can offer a standard analysis shrinks with every additional unit of uninterrupted off-drug time a design imposes, in exactly the pattern Section 3.2 traces mechanistically across the three designs. So the design implication drawn in Section 4.4, that designs with long off-drug windows carry the largest architecture-dependent risk, should be read as a caution for trials analyzed with the standard mixed-model methods of Section 2.1 specifically, not as a statement that latent-class biology makes long off-drug windows safe by construction; a class-aware analysis, where trial-relevant identifiability permits one, is not bound by the same limitation. For the same reason, the ordering of the three designs on architecture-sensitivity (Section 3.1) is a statement about how much power a fixed analysis model loses to the architecture choice, not a design-preference recommendation. At matched $N = 70$, covariance-moderation power under carryover happens to land close together for Hybrid (0.711) and CO (0.723) and well below both for OL+BDC (0.539), but the paths are allocated unevenly across the designs (18/18/17/17 for Hybrid's four paths versus 35/35 for CO's and OL+BDC's two), and design selection in practice turns on feasibility, blinding, and regulatory considerations that this power comparison alone does not capture. Investigators choosing among designs on carryover-robustness grounds should treat the Section 3.1 ordering as a hypothesis to examine under their own design's path allocation, not as a ranking to adopt directly.

5.2 4.2 Reporting recommendations

The choice of DGP architecture should be guided by the hypothesized biological mechanism, as summarized above. When the mechanism is ambiguous, as it often is in early biomarker development, we recommend that investigators run the simulation under both architectures

Criterion	Mean moderation	Covariance moderation
Biomarker role	Causal mediator	Statistical predictor
Mechanism	Deterministic scaling	Probabilistic association
Signal location	Mean structure	Covariance structure
Carryover impact	Modest (up to 7%)	Design-dependent, up to 31%
Appropriate when	Biomarker determines dose or PK	Biomarker indexes a latent subtype

and report the range of power estimates, treating covariance moderation as the conservative default because its larger carryover sensitivity keeps power estimates from being optimistic. A dual-channel sensitivity sweep across mixing weights is available in the companion paper[6]. Designing the trial to minimize carryover, through adequate washout periods, also reduces sensitivity to the architecture choice. More generally, simulation studies evaluating biomarker-moderated treatment effects should state explicitly the following.

- Whether the interaction is generated through mean moderation, covariance moderation, or both channels simultaneously.
- The biological rationale for the chosen mechanism.
- Whether any carryover model operates on the interaction signal consistently with that choice.
- When the mechanism is uncertain, sensitivity analyses under the alternative architecture.

These requirements specialize the Data-generating element of the ADEMP framework[8] to the class of architecture-sensitive biomarker simulations.

5.3 4.3 Possible mitigation strategies

Under covariance moderation, the power loss has two sources. The drug-exposure variable's variance shrinks (recoverable), and the biomarker-response correlation genuinely fades off drug (lost information no model can recover). Because off-drug points carry less signal, methods that down-weight or drop them may recover part of the loss. Candidates include restricting the analysis to on-drug points, weighting observations by expected residual drug effect (via the `weights` argument in `nlme::lme`), a within-subject contrast regressing each patient's on-minus-off mean difference on the biomarker, a two-stage random-slopes approach, or excluding the most contaminated early off-drug points. None of these strategies was tested by simulation here. A companion paper on carryover-mitigation strategies takes that up. We regard a systematic simulation comparison, analogous to the factorial comparison of Section 3, as important future work, and expect the on-drug-only restriction and the weighted analysis to be the most tractable candidates for a first empirical evaluation.

5.4 4.4 Relation to the wider literature

Almost every trial-simulation framework uses mean moderation. Simon[2], Freidlin and Korn[3], Renfro and Sargent[17], and standard N-of-1 simulation studies all put the biomarker effect in the mean. Even methodological work on N-of-1 carryover sharing authors with Hendrickson et al.[4] uses mean moderation, which suggests Hendrickson's differential-correlation choice was a one-off rather than a convention. The practical worry is that most published power estimates may be optimistic for biomarkers that really behave like covariance moderation, because they ignore the extra carryover-driven loss demonstrated here. The distinction matters most exactly in the designs precision medicine favors: treatment-switching designs (crossover, N-of-1 hybrid, basket/platform trials) put patients through off-drug phases where the covariance-moderation correlation decays, so the interaction signal in later periods is weaker than in the first. Parallel-group enrichment designs, with no off-drug phase, are immune, and give similar power under either architecture. Adaptive enrichment trials that use interim power calculations written in the mean-moderation idiom are at particular risk. If the truth is closer to covariance moderation and patients pass through washouts, the real power is lower than the rule assumes, which can trigger premature stopping or an over-narrow eligible population. This is a data-generating-process question, separate from, and not answered by, the analysis-model guidance in the PATH statement[7]. The architecture choice matters most in exactly the within-subject designs precision medicine finds most attractive, and least in the parallel-group designs where it is often, wrongly, assumed to be harmless.

5.5 4.5 Limitations

Several limitations qualify these results. The magnitude of the covariance-moderation loss in the focal Hybrid design (15.4%), and the largest value observed across the three designs, the 30.6% loss in OL+BDC, both fall below the 40-60% range suggested by earlier exploratory runs in this project, so the precise figures should be read as design- and calibration-specific rather than universal. A confirmatory run at $N = 140$ on the Hybrid design, analogous to the $N = 140$ OL+BDC check reported in an earlier version of this analysis, has not yet been re-executed under the current N-matching convention (see Reproducibility) and remains future work. The uniformly sub-nominal Type I error we observe reflects a known conservatism of the `corCAR1` Wald test that is common-mode across architectures but not itself resolved here. A cluster-robust remedy is characterized in a companion calibration study[11]. Nominal coverage was inferred from the correspondence between empirical and model-based standard errors rather than verified by a dedicated coverage simulation. The mitigation strategies of Section 4.3 are mechanistic hypotheses, not simulation-tested recommendations. Finally, the combined architecture that activates both channels at once is deferred entirely to a companion paper[6], and our results are matched to a single reference application (prazosin/PTSD) and three within-subject designs. Generalization to other biomarkers, designs, or effect sizes should be checked rather than assumed.

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6 5. Conclusions

1. The choice between direct mean moderation and covariance moderation for generating biomarker-treatment interactions in clinical trial simulations has substantive consequences for power estimates under carryover.

2. Covariance moderation is intuitively appealing whenever a biomarker is better understood as predictive than causal, and Section 4.1 sets out the reasons this is often the more defensible specification. That appeal carries a measurable cost under carryover, and the cost is design-dependent because it is driven by a mechanism (Section 3.2) that is itself design-dependent: carryover erodes the biomarker-response correlation that carries the covariance-moderation signal only during uninterrupted off-drug time, so the loss scales with how long each design sustains an off-drug period. In the focal Hybrid design, carryover reduces power by 15.4% under covariance moderation against 6.9% under mean moderation, roughly double. The crossover and open-label-plus-blinded-discontinuation contrast designs bracket this from opposite sides, from 3.0% (crossover, not distinguishable from zero, because within-subject AR(1) correlation already carries most of the signal) to 30.6% (OL+BDC, whose sustained blinded off-drug phase gives the correlation-erosion mechanism the most uninterrupted time to act).

3. The choice should be guided by the hypothesized biological mechanism. Mean moderation is appropriate for biomarkers that directly determine drug pharmacokinetics or pharmacodynamics, covariance moderation for biomarkers that statistically predict treatment response through latent subtype membership.

4. For the prazosin-PTSD application, with blood pressure as the predictive biomarker, covariance moderation is the more appropriate model, and the resulting power sensitivity to carryover should inform trial-design decisions about off-drug period duration.

5. The existing clinical trial simulation literature uses mean moderation almost exclusively. The Hendrickson et al. (2020) MVN approach is, to our knowledge, unique in the N-of-1 context. This means published power estimates for biomarker-moderated crossover and N-of-1 designs may be systematically optimistic for biomarkers that operate through a correlation-based mechanism.

6. Simulation studies for biomarker-moderated treatment effects should explicitly report their DGP architecture and, when the biological mechanism is uncertain, consider sensitivity analyses under the alternative.

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7 Reproducibility

N denotes the total number of participants across all randomization paths, and Section 3.1 runs every design at the same total, $N = 70$, allocated across that design's paths, so the cross-design comparison in Section 3 is a design comparison rather than a sample-size comparison. An earlier version of this analysis fixed the per-path count at 35 instead of the total, which gave the four-path Hybrid design 140 participants against 70 for the two-path designs and confounded any Hybrid advantage with twice the sample. Section 3.1 has been re-run at matched totals, and only the Hybrid figures changed. A confirmatory run at $N = 140$ on the Hybrid design, to check that the architecture-dependent ordering survives the power saturation that comes with a larger sample, is planned but has not yet been executed under the current N-matching convention and is not reported here. The Hendrickson et al. comparison arm of Section 2.2.4, including the vendored original source (with its one documented compatibility patch) and the positive-definiteness sweep, is at `analysis/scripts/quick-sim/hendrickson-original-comparison/`, with its output at `analysis/data/quick-sim/hendrickson-original-comparison/`. A structured, section-by-section summary of the argument, without the derivations, tables, or citations, is provided as a supplementary document (`bullets.Rmd`).

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